

Humidification

Final Lab Report
Unit Operations Lab 1
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1.Executive Summary (10 pts)

The objective of this experiment was to determine the heat and mass transfer coefficient of air–water contact in a wetted-wall humidification column to evaluate the interaction between the air flow rate and temperature to be incorporated in the design of cooling towers and humidifiers. Experimental procedures were carried out with three air flow rates (1, 3, and 5 SCFM) with both the ambient inlet air conditions as well as with heated inlet air conditions (55 °C & 70 °C) to analyze heat and mass transfer performance with the change in the Reynolds number.

The results showed that both the mass transfer coefficient k_y and heat transfer coefficient h increased with Reynolds number, confirming that greater turbulence enhances convective transport between air and the water film. However, while k_y increased steadily with velocity, h reached a maximum before decreasing at higher flow rates, indicating a trade-off between increased convection and reduced residence time. The number of transfer units (NTU) decreased, and the height of a transfer unit (HTU) increased with increasing air velocity and inlet temperature, consistent with a reduction in the overall driving forces for heat and mass transfer. The adjusted Sherwood number correlation followed the theoretical relationship ($Sh = 0.023 Re^{0.83} Sc^{0.44}$) within experimental uncertainty.

Overall, the experimental data agreed with humidification theory and established correlations, with small deviations attributed to measurement error and incomplete wetting. These findings confirm the expected dependence of heat and mass transfer on Reynolds number and provide practical insight into the design and optimization of humidification equipment.

2. Introduction (10 pts)

The main objective of this experiment was to determine the heat and mass transfer coefficients of air interacting with water in a wet wall humidification column. This experiment is important for understanding the transport processes that occur in many chemical engineering processes. Heat and mass transfer are key concepts that appear in almost every industrial system such as cooling towers, air humidifiers, and air conditioning units by studying how air flow rate and temperature affect these transfers, we can understand how to design and operate systems that rely on air water contact.

In this experiment, air was passed through a vertical column where water flowed down the inner wall. A centrifugal pump was used to circulate water from a reservoir through the column, while the airflow was provided from the utilities from the wall. The air was supplied at three different flow rates 1, 3 and 5 SCFM and under three temperature conditions: room temperature, moderate heat (about 55 °C), and high heat (about 70 °C). This gave us nine total wet runs to compare the effects of both flow rate and temperature. Steam and a heat exchanger were used to warm the air. The dials were used to control both the air and water flow, and thermocouples were placed along the column to measure temperature at different points. The humidity and temperature of the air exiting the column was also recorded using a steamometer.

The data collected from each trail were used to calculate dimensionless numbers such as Reynolds, Schimdt, Sherwood, and Lewis numbers. Which helps describe how heat and mass are transferred. Overall, this experiment helped demonstrate how changes in flow and temperature can affect both the rate and efficiency of heat and mass transfer humidification systems.

3. Theory (10 pts)

The main theories of this lab pertain to the heat and mass transfer within the column. There are many equations that cover heat and mass transfer, but the two below will be explored:

$$q = h * \Delta T \quad (1)$$

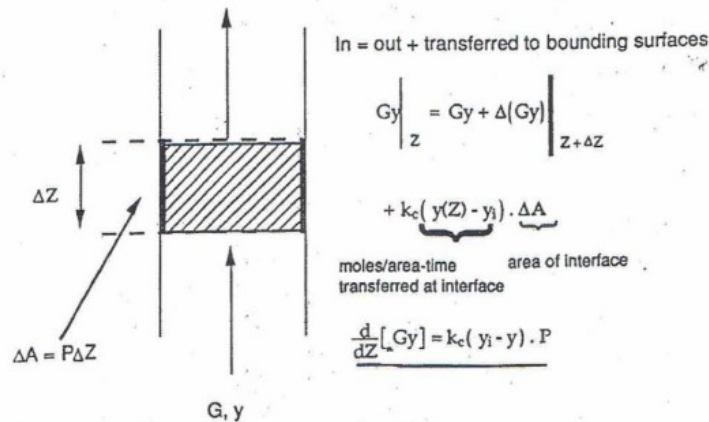
$$N_A = k_c * \Delta c_A \quad (2)$$

Equation (1) features a heat transfer equation, relating the amount of heat transferred, q , to the change in temperature, ΔT , and a heat transfer coefficient, h . Equation (2) features a mass transfer equation, relating the amount of moles transferred, N_A , to the concentration gradient, Δc_A , and a mass transfer coefficient, k_c . Deriving the coefficients for both equations takes a little bit of work and it must be applied specifically for humidification.

Mass Transfer Coefficient

The first step to find a mass transfer coefficient is conducting a mass balance on a differential height section of the column. This can be seen in Figure 1 from source 1:

Figure 1: Schematic of a Differential Element in the Column



The corresponding mass balance can be written in differential form as follows:

$$d(Gy) = k_c \cdot c \cdot (y_i - y) dA \quad (3)$$

- y_i and y refer to the mole fraction of water at the interface and in the gas, respectively
- G is the total gas flowrate (moles/time)
- c is the molar concentration of the gas, and dA is the area of the differential column height ($=Pdz$), where P is the perimeter of the tube and dz is a differential height)
- $d(Gy)$ is the differential change in the moles of water in the gas over the corresponding differential column height and is equal to t

This equation must be modified for our humidification purposes. The vapor fraction can be mapped as the mass fraction of water vs. dry gas rather than mole fraction. Additionally, the left-hand term of equation (3) can be rewritten as follows:

$$Y' = \frac{\text{mass water}}{\text{mass dry gas}} = \frac{\frac{y}{1-y} * M W_{H_2O}}{M W_{AIR}} \quad (4)$$

$$d(Gy) = \frac{G'_s}{M W_{Air}} dY' \quad (5)$$

Which makes the differential equation:

$$G'_s dY' = k_Y (Y'_i - Y') P dz \quad (6)$$

Where k_Y is the mass transfer coefficient defined in terms of a mass ratio driving force. It can be derived from k_c through the equation:

$$k_Y = k_c (M W_{Air}) * c * (1 - y)_M \quad (7)$$

And:

$$(1 - y)_M = \frac{y_1 - y_2}{\ln \left(\frac{(1 - y)_2}{(1 - y)_1} \right)} \quad (8)$$

Eventually, this equation comes into the form of the following:

$$\ln \left(\frac{Y'_i - Y'_1}{Y'_i - Y'_2} \right) = \frac{k_Y P Z}{G_S} \quad (9)$$

- Where P is the perimeter (m)
- Z is the height of the column (m)
- G_S is the flowrate of water (kg/s)
- k_Y is the mass transfer coefficient (kg/s)
- Y'_i is the vapor fraction at the interface
- Y'_1 is the vapor fraction at the inlet
- Y'_2 is the vapor fraction at the outlet

This form is found from integrating equation (6).

Heat Transfer Coefficient

Likewise with the mass transfer coefficient, a balance is made on the differential height section of the column. The equation becomes:

$$d(G C_p T) = h (T_i - T) dA \quad (10)$$

Where C_p is the molar heat capacity of air and G is the mass flow rate of air. In a similar fashion to the mass transfer coefficient, a lot of the terms in this equation can be rewritten:

$$d(G C_p T) = G'_S C'_S dT \quad (11)$$

Where G'_S is the mass flowrate of dry air and C'_S is the heat capacity of the gas and vapor mixture. This is found through:

$$C'_S = C'_{Air} + Y' * C'_{H_2O} \quad (12)$$

Where C'_{Air} is the heat capacity of dry air and C'_{H_2O} is the heat capacity of water vapor. To use equation (11), one must note that C'_S is a function of Y' , or $C'_S = C'_S(Y')$. However,

it can be assumed that Y' will not vary appreciably across the column, meaning that C'_s is constant. Now equation (11) can be integrated to yield its final form:

$$\ln \left(\frac{T_i - T_1}{T_i - T_2} \right) = \frac{hPZ}{G_s C_s} \quad (13)$$

This lab will utilize equations (9) and (13) to calculate the mass and heat transfer coefficients respectively. Further calculations will be conducted to find rheological and dimensionless properties.

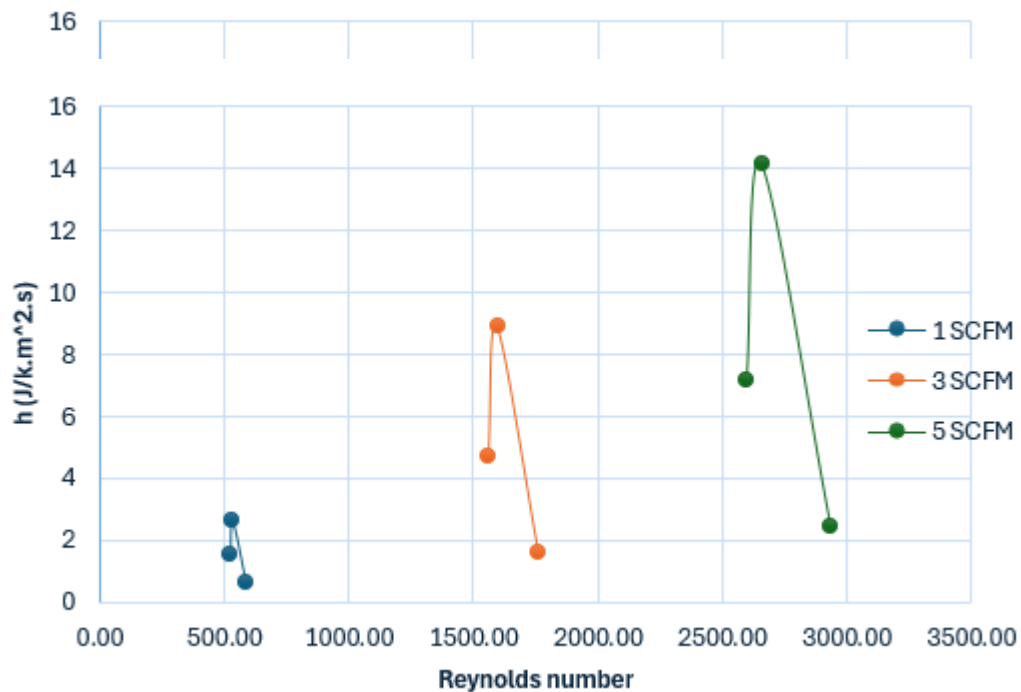
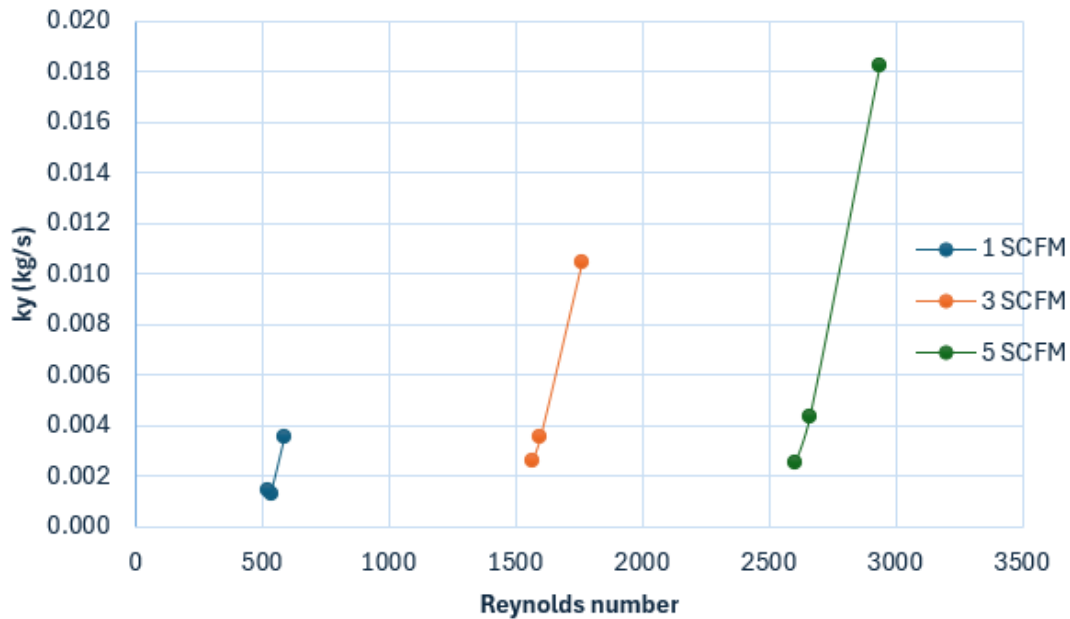
Table 1: Units

QUANTITY	UNITS
Perimeter (P)	Meters, (m)
Column Length (z)	Meters, (m)
Flowrate of Water (G_s)	$\left(\frac{kg}{s} \right)$
Flowrate of Air (G'_s)	$\left(\frac{kg}{s} \right)$
Mass Transfer Coefficient (k_y)	$\left(\frac{kg}{s} \right)$
Heat Transfer Coefficient (h)	$\left(\frac{J}{K * m^2 * s} \right)$
Temperature (T)	($^{\circ}C$)

4. Results (15 pts)

The experimental data were analyzed to evaluate how air flow rate and temperature influenced the heat and mass transfer performance of the humidification column. Both the heat and mass transfer coefficients increased with Reynolds number across all flow rates, reflecting the enhanced convective transport that occurs at higher air velocities. At lower flow rates, the weaker turbulence led to thicker boundary layers and reduced transfer between the air and water film. As the flow increased, these boundary layers became thinner, resulting in higher coefficients and improved mixing along the column. However, while the mass transfer coefficient (k_y) continued to increase steadily with Reynolds number, the heat transfer coefficient (h) reached a maximum before decreasing at higher flow rates, forming a negative-parabolic trend. This suggests that although increased turbulence initially improves heat transfer, the reduced residence time of air at very high velocities limits the total energy exchanged. The differing behaviors indicate that mass transfer in this system is more strongly influenced by turbulence, whereas heat transfer is partially constrained by shorter contact times and weaker temperature gradients at higher flow rates. Overall, greater airflow promoted convective transport, with the largest average coefficients still occurring at the highest Reynolds numbers. Mass and heat transfer coefficients were graphed as a function of inlet air temperature, with inlet air flow rate as a parameter. The relationships discussed for mass and heat transfer coefficients can be found in figures 2 and 3.

Figure 2: Mass Transfer Coefficient vs Reynolds Number



In contrast, the number of transfer units (NTU) decreased while the height of a transfer unit (HTU) increased with increasing Reynolds number and inlet air temperature. Although higher velocities enhanced individual heat and mass transfer coefficients, the corresponding temperature rise weakened the driving potential for both heat and mass transfer. This reduced gradient led to lower NTU values and therefore higher HTU values, indicating a less efficient overall transfer per unit column height. The 1 SCFM trials showed the highest NTU and lowest HTU, suggesting that slower airflows allow

more time for air–water equilibrium to be reached. These observations align with humidification theory, where increased velocity and temperature strengthen convective effects but simultaneously diminish the net driving forces that control the rate of transfer. Like before, NTU and HTU were graphed as a function of inlet air temperature. The relationship of NTU and HTU to Reynolds number can be seen in figure 4 and 5.

Figure 4: Number of Transfer Units (NTU) vs Reynolds Number

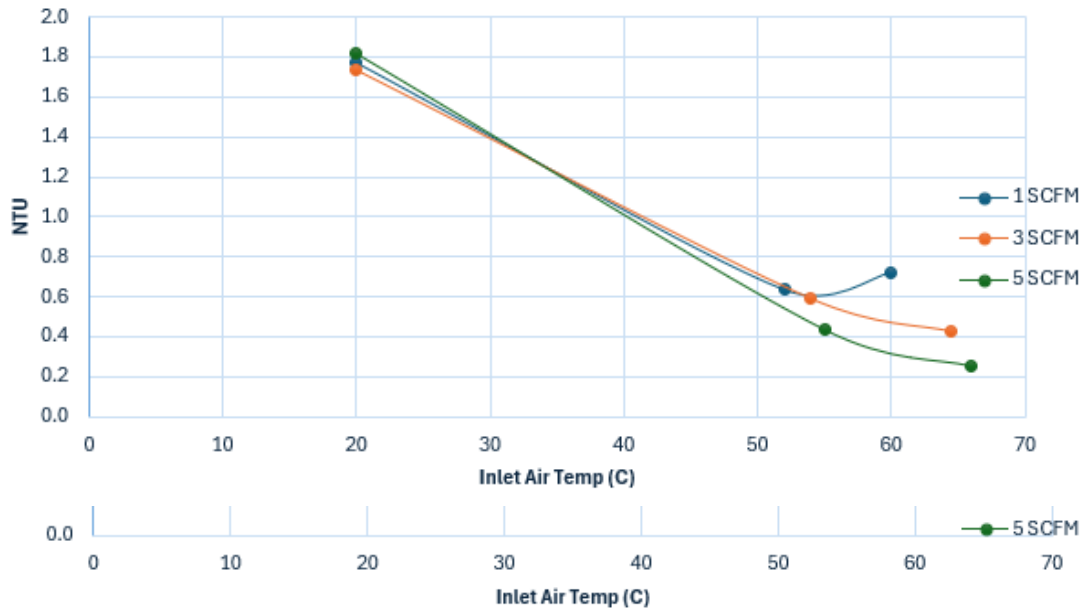
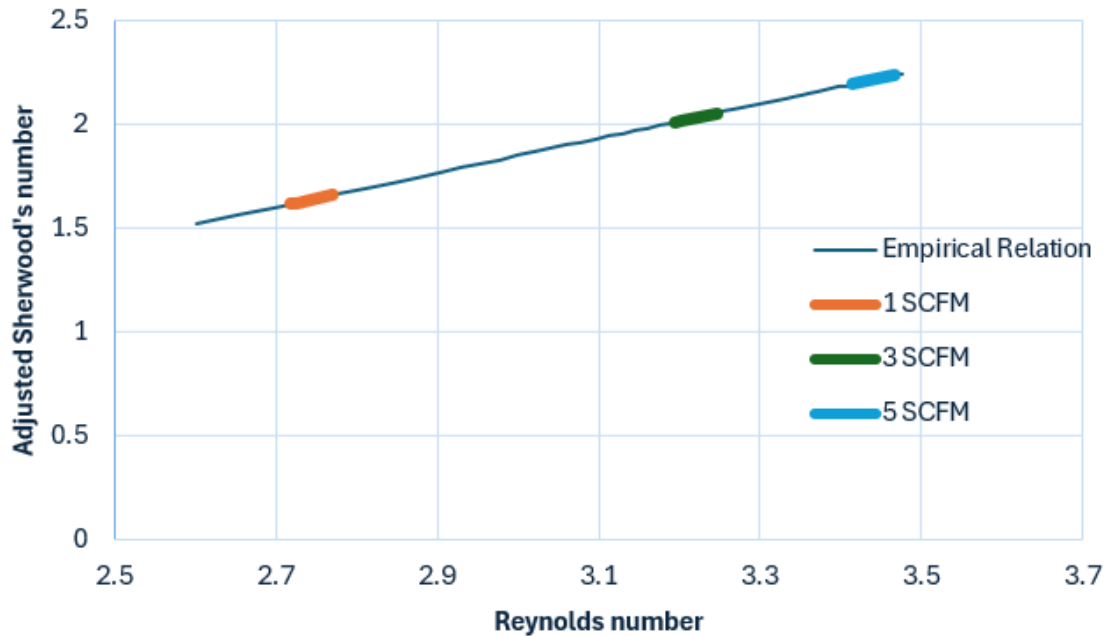


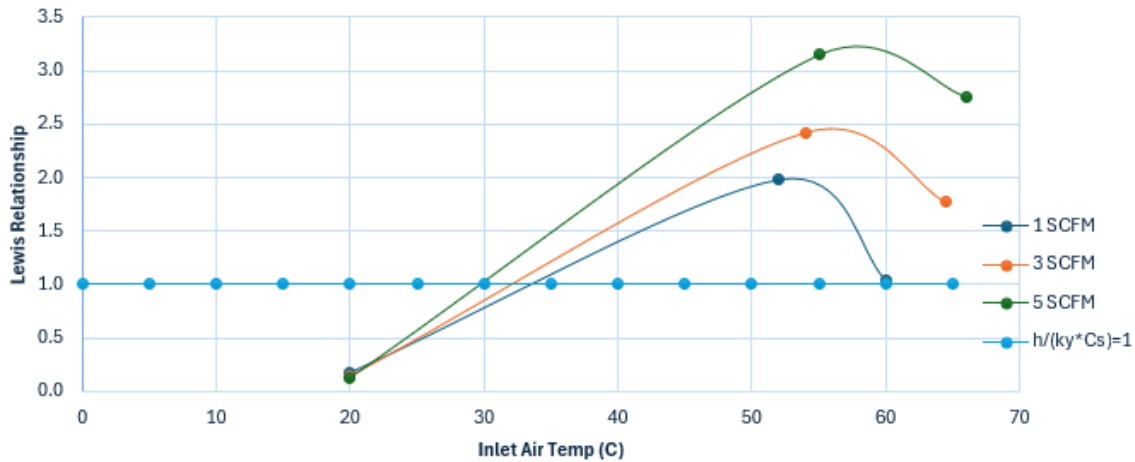
Figure 5: Height of Transfer Units (HTU) vs Reynolds Number

The adjusted Sherwood number, defined as the Sherwood number divided by $Sc^{0.44}$, was plotted against the Reynolds number on a log–log scale to compare the experimental results with the empirical correlation $Sh = 0.023 Re^{0.83}$, as can be seen in figure 6. The empirical relation (shown as the long blue line) represents expected behavior for air–water systems in wetted-wall columns, while the shorter colored segments correspond to the experimental data collected at 1, 3, and 5 SCFM. The measured points follow the same general slope as the correlation, indicating that the dependence of mass transfer on flow regime was captured correctly. However, the experimental data lie slightly below the theoretical curve, suggesting that the observed mass transfer rates were somewhat lower than predicted, which is likely due to temperature fluctuations, small variations in humidity measurement, or incomplete wetting of the column surface. Despite these minor deviations, the overall agreement confirms that the experimental system follows the expected Reynolds–Sherwood relationship within reasonable experimental uncertainty.

Figure 6: Adjusted Sherwood Number vs Reynolds Number



The Lewis relationship, defined as $Le = \frac{h}{k_y C_s}$, was plotted against inlet air temperature for each flow rate, with a reference line at $Le = 1$, representing the ideal Lewis relationship. This can be seen in figure 7. At lower temperatures (around 20 °C), all trials showed $Le < 1$, indicating that the measured heat transfer coefficients were smaller than those predicted by the Lewis relation. As inlet air temperature increased, Le rose for all flow rates, reaching maximum values between approximately 2 and 3 near 55–60 °C. This suggests that higher temperatures enhanced convective heat transfer more strongly than mass transfer, likely due to increased air turbulence and thermal diffusivity. However, beyond 60 °C, Le began to decline slightly, implying that very high inlet temperatures reduced the effective temperature gradient and residence time, limiting overall heat transfer. Among the trials, the 5 SCFM run showed the most pronounced deviation above unity, consistent with greater turbulence effects at higher velocities. Overall, the results indicate that the Lewis relationship holds approximately at lower temperatures and moderate flow rates but diverges at higher temperatures and velocities, where heat and mass transfer no longer scale proportionally.

Figure 7: Lewis Relationship vs Inlet Air Temperature ($^{\circ}\text{C}$)

Overall, the experimental results demonstrated the expected qualitative trends for a humidification column, where increasing air flow rate enhances convective heat and mass transfer but reduces the overall transfer effectiveness due to diminished driving forces. Both k_y and h increased with Reynolds number, confirming stronger turbulence and boundary-layer disruption at higher flow rates. Conversely, NTU decreased and HTU increased with higher velocities and inlet air temperatures, indicating reduced efficiency per unit column height. The adjusted Sherwood number results followed the theoretical correlation within experimental uncertainty, supporting the validity of the measured mass-transfer behavior. While the calculated Lewis relationship deviated quite significantly than the expected results, it was still within experimental error ranges. Together, these findings confirm that while elevated flow rates improve local transport coefficients, they also lessen the column's overall approach to equilibrium.

5. Discussion (15 pts)

The experimental results obtained from the study generally align with the theoretical behavior expected for simultaneous heat and mass transfer in a wetted wall humidification column. Both the mass transfer coefficient k_y and the heat transfer coefficient h increased with Reynolds number, as shown in figures 1 and 2, indicating that higher air velocities enhance convective transport between the gas and liquid film. This agrees with the theoretical correlation for air–water systems, $Sh = 0.023 Re^{0.83} Sc^{0.44}$, provided in the lab manual [1]. Increasing turbulence at higher flow rates thins the boundary layers and promotes greater interaction between air and the water film, which increases both coefficients.

While k_y increased monotonically, displayed a negative-parabolic trend at higher Reynolds numbers (Figure 2). This behavior suggests that, although turbulence initially improves convective heat transfer, the shorter residence time of air at very high velocities limits total energy exchange. Thus, the system exhibits a trade-off between increased convection and reduced contact time, leading to a maximum in h followed by a gradual decline.

The variation of NTU and HTU with Reynolds number, shown in figures 3 and 4, also follows theoretical expectations. As air velocity and inlet air temperature increased, NTU decreased while HTU increased. This inverse relationship results from a reduction in the overall temperature and humidity driving forces at higher flow rates. Even though higher turbulence improves local transfer coefficients, the air approaches saturation more rapidly, leaving less potential for additional transfer per unit column height. The lowest air flowrate (1 SCFM) therefore produced the highest NTU and lowest HTU, confirming that slower flows allow more complete approach to equilibrium. These results correspond closely to equations (12) and (13) in the manual, which relate NTU and HTU to the flow rate and driving forces across the column.

The adjusted Sherwood number correlation in figure 5 further validates the experimental data. The measured points at 1, 3, and 5 SCFM followed the same slope as the theoretical line, confirming that the dependence of mass transfer on flow regime was correctly captured. However, the experimental data lie slightly below the predicted correlation, indicating that the actual mass transfer rates were somewhat lower than ideal. This deviation likely arises from incomplete column wetting, temperature fluctuations, or instrumental uncertainty in humidity measurement each of which can cause significant variation in vapor mass-fraction calculations.

Potential sources of experimental error include:

- Incomplete water film coverage along the column wall, which decreases the effective contact area and underestimates $k_y \wedge h$.
- Humidity sensor drift or lag in the wet-bulb response, which could introduce scatter in Y' calculations
- Non-uniform film temperature caused by minor heat losses to the surroundings, violating the adiabatic assumption.

-Flow-meter uncertainty, affecting calculated Reynolds numbers and derived dimensionless groups.

To minimize these issues, future runs could incorporate a more uniform liquid distributor at the top of the column, insulation around the test section, and digital humidity logging for higher precision.

The assumptions made steady-state operation, negligible heat loss, dilute vapor mixture, and uniform film-remain mostly valid but introduce modest error where violated. Despite these limitations, the qualitative trends across all figures confirm the reliability of the collected data.

In Summary, Figures 1-5 collectively demonstrate that higher air flow enhances local convective heat and mass transfer while simultaneously reducing the overall transfer effectiveness. The experiment verified the Reynolds-Sherwood relationship. Quantitatively, results were slightly below literature values, primarily due to systematic experimental limitations. These findings reinforce the theoretical understanding that increasing flow strengthens turbulence and coefficient magnitude but shortens residence time, an essential consideration for designing humidification towers and air-water contact equipment. Future work could extend the air-temperature range, automate data collection, and compare results with alternate column geometries to improve correlation accuracy.

6. References (10 pts)

- [1] Department of Chemical Engineering, *Humidification Lab Manual*, University of Illinois at Chicago, 2025.
- [2] McCabe, W.L., Smith, J.C., Harriott, “Unit Operations in Chemical Engineering” 7th Edition, McGraw-Hill, New York.
- [3] Humidity calculator (Quadco Engineering): Relative humidity to mass fraction H₂O
<https://www.quadco.engineering/en/know-how/cfd-calculate-water-fraction-humid-air.htm>
- [4] Engineering Toolbox. “Air – Dynamic and Kinematic Viscosity.” The Engineering Toolbox. https://www.engineeringtoolbox.com/air-absolute-kinematic-viscosity-d_601.html
- [5] University of Michigan. “Chapter 11 — Example 11.1: Prediction of Binary Gas Diffusivities.” Elements Course Lectures.
<https://websites.umich.edu/~elements/course/lectures/eleven/exam4.htm>

7. Appendix I: Data Tabulation/Graphs (10 pts)

For each temperature setting, the column was operated for three dry runs and three wet runs, giving a total of 18 trials. The dry runs were used to determine the inlet air's water mass fraction, which were then applied to calculate key properties for the wet runs. During all trials, the temperatures along the column, inlet and outlet air temperatures, wet- and dry-bulb temperatures, and outlet air relative humidity were recorded. The calculations shown in table 3 and table 4 are explained in more detail in the theory, results, and sample calculations sections.

Table 2 – Raw Data

trial #	wettness	temp setting	air flow (SCFM)	water flow (G.P.)	inlet air temp (C)	Thermocouple Readings						steamometer	
						T1 (C)	T2 (C)	T3 @	T4 @	T5 @	T6 @	T7 (F)	Humidity (%)
1	dry	1	1	0	20	21.4	22.1	22.1	22.1	22.1	21.8	72.3	8.2
2	dry	1	3	0	20	21.3	22.1	22.1	22.1	22.1	21.8	71.6	6.5
3	dry	1	5	0	20	21.1	22	22.1	22.1	22.1	21.8	71	5.4
4	wet	1	1	0.163	20	22.7	23.6	26.6	25.9	25.8	25.4	73.9	69.8
5	wet	1	3	0.163	20	23.1	23.9	27.3	26.5	26.2	25.7	74.6	66.7
6	wet	1	5	0.163	20	23.2	23.9	27.2	26.4	26	25.4	74.8	62.2
7	dry	2	1	0	58	31	40.5	27.4	27.9	31.1	24.8	98.6	5.5
8	dry	2	3	0	59	31.8	42.3	28.4	28.8	32.2	24.7	102	5.1
9	dry	2	5	0	60	33.7	45.4	30.2	30.3	34.1	24.5	108.1	5.1
10	wet	2	1	0.163	52	31.6	37.4	29.8	29.1	29	28.9	97.3	43
11	wet	2	3	0.163	54	31.9	38.7	30.3	29.6	29.4	29.1	99.1	40.1
12	wet	2	5	0.163	55	32.6	40.3	30.6	29.8	29.5	29.2	101.6	36.3
13	dry	3	1	0	69	41.3	51.3	45.2	45.6	46.2	36.4	103.2	4.2
14	dry	3	3	0	70	41.9	52.9	45.5	46.4	47.7	35.2	107.2	3.9
15	dry	3	5	0	70	43.4	54.8	46.6	48.3	49.1	35.4	112.1	3.7
16	wet	3	1	0.163	60	33.7	41.1	33	32	31.9	31.9	101.6	46.9
17	wet	3	3	0.163	64.5	34.9	43.8	32.9	32	31.5	31.6	106.3	38.8
18	wet	3	5	0.163	66	36.3	46.2	32.6	31.7	31.5	31.3	110.4	32.5

Table 3 – Mass Transfer Calculations

trial #	Ti @	T7 @	Yi (kg water/kgair)	Y1 (kg water/kgair)	Y2 (kg water/kgair)	Gs (kg/s)	ky (kg/s)	NTU	HTU
4	26.1	23.28	0.0146	0.0012	0.0123	0.00058	0.00356	1.774	0.749
5	26.67	23.67	0.0145	0.0009	0.0121	0.00173	0.01045	1.737	0.765
6	26.53	23.78	0.0134	0.0008	0.0113	0.00287	0.01823	1.819	0.730
10	29.3	36.28	0.0161	0.0062	0.0108	0.00057	0.00127	0.636	2.087
11	29.77	37.28	0.0158	0.0060	0.0104	0.00172	0.00355	0.591	2.247
12	29.97	38.67	0.0155	0.0063	0.0095	0.00287	0.00434	0.434	3.060
16	32.3	38.67	0.0200	0.0077	0.0141	0.00058	0.00145	0.721	1.842
17	32.13	41.28	0.0190	0.0075	0.0115	0.00172	0.00256	0.426	3.118
18	31.93	43.56	0.0180	0.0071	0.0095	0.00287	0.00250	0.250	5.307

Table 4 – Heat Transfer and Dimensionless Quantities Calculations

trial #	Cs (J/kg.K)	h (J/km ² .s)	Lewis relationship	superficial V (m/s)	Reynolds number	diffusion coefficient (m ² /s)	Schmidt #	Sherwood #
4	1032.3	0.637	0.173	0.129	587.9	2.51E-05	0.6006	36.55
5	1031.9	1.577	0.146	0.386	1763.3	2.52E-05	0.5987	90.81
6	1029.9	2.433	0.130	0.643	2936.6	2.51E-05	0.5991	138.72
10	1035.0	2.610	1.978	0.129	532.4	2.55E-05	0.6500	34.85
11	1034.5	8.899	2.423	0.386	1596.5	2.56E-05	0.6483	86.60
12	1033.8	14.137	3.149	0.642	2658.5	2.56E-05	0.6475	132.17
16	1042.3	1.565	1.036	0.129	522.4	2.60E-05	0.6533	34.38
17	1040.5	4.720	1.771	0.386	1563.1	2.60E-05	0.6539	85.42
18	1038.5	7.170	2.758	0.642	2600.0	2.59E-05	0.6547	130.38

Table 5 – Additional parameters

Length (m)	1.328		temp setting	air viscosity (Pa.s)
radius (m)	0.0344		no temp	1.81E-05
diameter (m)	0.0687		low temp	2.00E-05
Perimeter (m)	0.2159		high temp	2.04E-05
air density (kg/m ³)	1.204			
air pressure (psi)	60			
flow (SCFM)	mdry (kg/s)			
1	0.000568			
3	0.001705			
5	0.002841			

8.

Appendix II: Error Analysis (10 pts)

There were many instances where error could arise in this lab. Firstly, there are 9 measurements conducted during each trial and 4 other parameters that are adjusted for each type of trial. Accidentally reading different measurements or setting parameters to different settings could result in error.

The largest portion of the measurements was temperature, which was read from 6 different thermocouples. In this system, the thermocouples read temperature values at different stages of the column, which allow us to analyze the heat and mass transfer coefficients. Thermocouples generally have an error of around $\pm 0.5\%$, especially when they have not been recently calibrated. No such calibration was conducted for this lab, and thus an error of $\pm 0.5\%$ is accepted for the readings. An error as small as this could have serious implications across 6 readings that are all utilized for calculations- a cascading effect. Other than equipment failure, there are many instances where human error could have occurred. Most notably, error could occur when one reads the thermocouples before the system has reached “steady state”. There were multiple instances where a thermocouple reading would jump 0.1°C as it was being recorded, meaning that the system was still heating up. Steady state is hard to reach in this system, as the water recirculating will continue to heat up slowly throughout time. However, not waiting for the inlet air temperature to reach a steady temperature when the heat exchanger is on is a definite and large source of error, as the temperature profile of the hot air will not have fully developed yet, leading to improper readings from the thermocouple. Lastly, one temperature reading was taken from the Steamometer, which would frequently shut off during the lab and need to be turned back on. Not letting the Steamometer come to equilibrium is another source of error for both temperature and relative humidity readings. NOTE: Because trials were never repeated for the same conditions, we cannot conduct a variability calculation.

Another source of error is the accuracy of air flow. The gauge only reads values starting from 3 SCFM but has a maximum flowrate of 5 SCFM. To create 3 conditions, a low, medium, and high, 1 SCFM was estimated on the gauge, which may not have been consistent for every run, as there is no line to demarcate the amount. Additionally, the

marker would float around and vary above 4 SCFM, making it hard to set it consistently at the 5 SCFM for our trials.

To observe the effects of error, let's examine the Dry Runs in the column. First looking at the dry runs at room temperature (low):

$$\text{Trial 1 (1 SCFM)} = 8.2 \% RH$$

$$\text{Trial 2 (3 SCFM)} = 6.5 \% RH$$

$$\text{Trial 3 (5 SCFM)} = 5.4 \% RH$$

$$(\%Difference)_{2,1} = \frac{8.2 - 6.5}{8.2} * 100 = 20.73 \%$$

$$(\%Difference)_{3,2} = \frac{6.5 - 5.4}{6.5} * 100 = 16.92 \%$$

Now looking at the dry runs at temperature setting 2 (medium):

$$\text{Trial 7 (1 SCFM)} = 5.5 \% RH$$

$$\text{Trial 8 (3 SCFM)} = 5.1 \% RH$$

$$\text{Trial 9 (5 SCFM)} = 5.1 \% RH$$

$$(\%Difference)_{8,7} = \frac{5.5 - 5.1}{5.1} * 100 = 7.84 \%$$

$$(\%Difference)_{9,8} = \frac{5.1 - 5.1}{5.1} * 100 = 0.00 \%$$

Lastly looking at the dry runs at temperature setting 3 (high):

$$\text{Trial 13 (1 SCFM)} = 4.2 \% RH$$

$$\text{Trial 14 (3 SCFM)} = 3.9 \% RH$$

$$\text{Trial 15 (5 SCFM)} = 3.7 \% RH$$

$$(\%Difference)_{14,13} = \frac{4.2 - 3.9}{4.2} * 100 = 7.14 \%$$

$$(\%Difference)_{15,14} = \frac{3.9 - 3.7}{3.9} * 100 = 5.128 \%$$

There is a clear distinction in the relationship between relative humidity and air flowrate for the medium temperature setting. Here we can see that RH was equal for 3 SCFM and 5 SCFM air flowrates, which generally should not be correct. An error in either recording or setting our flowrates is observed, and the data can lead to an improper calculation of the mass transfer coefficient, k_y . An incorrect calculation of k_y may have cascaded to an incorrect calculation for the Lewis Relationship, $Le = \frac{h}{k_y C_s}$, as we observed that it was

unideal ($\phi \neq 1$) for all flowrates of air. This stresses the importance of minimizing human error and strategically choosing the settings in which to run our experiments.

9. Appendix III: Sample Calculations (10 pts)

Mass Transfer

Step 1:

Humidification Experimental Data:

Air flow rate: 1 SCFM

Water flow rate: 0.163 GPM

Inlet air temperature: 26.1 °C

Outlet air temperature: 23.3 °C

Relative humidity (inlet): 8.2 %

Pressure: 101325 Pa

Humidity Ratios (Y_1', Y_i', Y_2') The humidity mass fractions were obtained using the Quadco Engineering Humidity Calculator [3]. The calculator requires inputs of temperature, relative humidity, and total pressure (entered in Pascals)

Y_i' , Y_1' , and Y_2' are the mass fractions of water with respect to air at the interface, inlet, and outlet, respectively. Y_i' is calculated at the saturation temperature corresponding to the average temperature of the packing (interface). Y_1' and Y_2' are obtained using the Quadco Engineering Humidity Calculator, which takes the air temperature, relative humidity, and total pressure as inputs.

For Trial 4, the following values were determined:

$Y_i' = 0.0146$ kg water/kg air

$Y_1' = 0.0012$ kg water/kg air

$Y_2' = 0.0123$ kg water/kg air

Step 2:

Now we can get the number of transfer units since we have our mass ratios

$$Ntu = \ln \left(\frac{Y' - Y_1}{Y' - Y_2} \right)$$

$$Ntu = \ln \left(\frac{0.0146 - 0.0012}{0.0146 - 0.0123} \right) = 1.774$$

The Height of a Transfer Unit HTU was found from the relationship $HTU = \frac{z}{NTU}$. For a column packing height of 1.329 m, $HTU = \frac{1.329}{1.774} = 0.749 \text{ m}$

The overall gas-phase mass transfer coefficient (k_y) can be determined directly from the logarithmic humidity difference using:

$$k_y = \frac{G_s \cdot \ln \left(\frac{Y_1' - Y_1'}{Y_1' - Y_2'} \right)}{Pz}$$

Known values:

$$Y_1 = 0.0146, Y_2 = 0.0012, Y_1' = 0.0123, G_s = 0.00058 \text{ kg/s}, P = 0.24 \text{ m}, z = 1.329 \text{ m}$$

Mass Flow Rate of dry air is found using the following equation, where Q_{SCFM} is the flow rate of air:

$$m_{dry} = Q_{SCFM} \left(\frac{0.0283168}{60} \right) (1.204) = Q_{SCFM} \cdot 5.638 \cdot 10^{-4} \frac{\text{kg}}{\text{s}} (\text{dry air})$$

Since the flow rate of air is 1 SCFM for this trial, it's equal to $m_{dry} = 5.638 \cdot 10^{-4} \frac{\text{kg}}{\text{s}}$.

This flow rate is used to determine the mass flow rate of wet air using water mass fraction Y_i . When the air contains moisture the total mass flow rate of the gas mixture is obtained from,

$$G_s = \frac{m_{dry}}{1 - Y_i} = \frac{5.638 \cdot 10^{-4}}{1 - 0.0146} = 0.0058 \text{ kg/sec}$$

This accounts for the water vapor content in the air. G_s indicates the rate at which the moist air mixture moves through the column and affects the overall mass transfer process. Mass Velocity in this step determines the mass velocity of the moist air stream, which is the mass flow rate per unit cross-sectional area of the column.

Thus,

$$k_y = \frac{0.00058 \cdot \ln \left(\frac{0.0146 - 0.0012}{0.0146 - 0.0123} \right)}{(0.24)(1.329)} = 0.00356 \text{ kg/s}$$

Heat Transfer

For Heat Transfer we will use the equation. First let's solve $C_s = C_{air} + Y_i' C_{water}$

$$h = \left(\frac{G_s \cdot C_s}{P \cdot z} \right) \cdot \ln \left(\frac{T_i - T_1}{T_i - T_2} \right)$$

$$C_s = C_{air} + Y_i \cdot \text{water} C_s$$

In this step we will calculate the Gas Heat Capacity (CS)

$$C_s = 1005 + (0.0146)(1864) = 1032.2 \text{ J / kg} \cdot \text{K}$$

Now we will compute the logarithmic temperature term

$$\ln \left(\frac{T_i - T_1}{T_i - T_2} \right)$$

$$\ln \left(\frac{26.1 - 20.0}{26.1 - 23.28} \right) = 0.772$$

After computing the logarithmic temperature term, we now substitute into equation for h,

$$h = \left(\frac{G_s \cdot C_s}{P \cdot z} \right) \cdot \ln \left(\frac{T_i - T_1}{T_i - T_2} \right) \quad h = \left(\frac{0.00058 \cdot 1032.2}{0.216 \cdot 1.329} \right) \cdot \ln \left(\frac{26.1 - 20.0}{26.1 - 23.28} \right)$$

$$h = 0.637 \frac{\text{J}}{\text{K} \cdot \text{m}^2 \cdot \text{s}}$$

Now we will solve the Reynolds number we need ρ (air density), μ (dynamic viscosity), column diameter d , and superficial velocity u where:

density of air = 1.18 kg/m³

column diameter = 0.069m

superficial velocity = 0.129 m/s

dynamic viscosity of air = 1.84×10^{-5} Pa·s

$$\Re = \frac{\rho d u}{\mu}$$

$$\Re = \frac{(1.18)(0.069)(0.129)}{1.84 \cdot 10^{-5}} = 588$$

The Schmidt number (Sc) is a dimensionless parameter that describes the ratio of momentum diffusivity (viscosity) to mass diffusivity. It provides insight into how momentum and mass are transported in a fluid. The equation is given by

$$Sc = \frac{\mu}{\rho \cdot D_{AB}}$$

Using properties at 20–23 °C and 1 atm, the needed properties are:

- Dynamic viscosity was found using a viscosity calculator [4], where $\mu = 1.84 \times 10^{-5}$ for this trial.
- Air density, $\rho = 1.204 \text{ kg/m}^3$.
- To find diffusivity of water vapor in air D_{AB} , we use the fuller-Schettler-Giddings correlation [5]. For the first trial, it is:

$$D_{AB} = 0.001 \frac{T^{1.75}}{P \left[\left(\sum V_A^{\frac{1}{3}} + \sum V_B^{\frac{1}{3}} \right)^2 \right]} \left(\frac{1}{M_A} + \frac{1}{M_B} \right) =$$

$$D_{AB} = \frac{0.001 \frac{((26.1 + 273.15) \text{ K})^{1.75}}{1 \text{ atm} \left[\left(13.1^{\frac{1}{3}} + 20.1^{\frac{1}{3}} \right)^2 \right]} \left(\frac{1}{\frac{18.015 \text{ g}}{\text{mol}}} + \frac{1}{\frac{28.97 \text{ g}}{\text{mol}}} \right)^{\frac{1}{2}}}{\left(\frac{100 \text{ cm}}{1 \text{ m}} \right)^2} = 2.51 \times 10^{-5} \frac{\text{m}^2}{\text{s}}$$

In this calculation, P represents the pressure in atmospheres (atm), and $\sum V$ the sum of the diffusion volumes for each species, which are dimensionless values obtained from Source 4. M refers to the molar mass of each species, where species A is water vapor and species B is air. T is the temperature across the column (T_i) measured in Kelvin.

The diffusion coefficient D_{AB} is originally given in cm^2/s , so it was converted to m^2/s to keep the units consistent throughout the calculations.

Thus, the Schmidt number for this trial is:

$$Sc = \frac{1.84 \times 10^{-5}}{(1.204)(2.5 \cdot 10^{-5})} = 0.63$$

Step 7:

The Sherwood number is evaluated from the correlation

$$Sh = 0.23 (Re)^{0.83} (Sc)^{0.44}$$

$$Sh = 0.23 (588)^{0.83} \cdot (0.63)^{0.44} = 36.55$$

This dimensionless number represents the ratio of convective mass transfer to diffusive mass transfer.

The adjusted Sherwood number is calculated using the empirical correlation

$$Sh_{adj} = 0.23 Re^{0.83}$$

$$Sh_{adj} = 0.23 (588)^{0.83} = 35.6$$

This correlation is applied after determining the Sherwood number and is commonly used to plot Sherwood number against Reynolds number as part of empirical validation.

Finally, The Lewis relationship was calculated to compare the experimentally determined heat and mass transfer coefficients and verify whether the humidification process followed the expected theoretical correlation between heat and mass transfer. For this trial,

$$Le = \frac{h}{k_y C_s} = \frac{0.637}{0.00356 * 1032.2} = 0.173$$

Although the calculated Lewis number deviated significantly from 1, it remains within acceptable experimental error due to measurement uncertainties and equipment limitations.

This concludes all sample calculations.

10. Appendix IV: Individual Team Contributions

	Time (Hours)			
	Joshua Olusoji	Omar Zaruk	Sanad Abu Hweij	Qussay Suleiman
Operator (Both Lab Days)	4	4	4	3
Pre-Lab Editing	1	1	1	2
Experimental Objective (Prelab)				1
Apparatus (prelab)	3			0
Materials & Supplies (prelab)			1	0
Procedure (prelab)		3		0
Project Safety Evaluation (prelab)				2
Final Lab Editing	1	1		1
Executive Summary (final lab)		1		0
Introduction (Final lab)			3	0
Theory (Final Lab)	2			0
Results (final lab)		1		3
Discussion (final lab)		4		0
References (final and/or prelab)	1			0.5
Data Tabulation/Graphs (final)				1
Error Analysis (final lab)	2			0
Sample Calculations (final Lab)			7	0
Power Point Presentation	1		1	2

Total	15	15	17	15.5
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